

Evaluation Metrics

Objective

In this assignment, you will train a Logistic Regression model on the famous Iris dataset. Your goal is not just to build the model, but to critically evaluate its performance using the classification metrics we covered in class.

Instructions

1. Data Preparation

- Import the dataset using `sklearn.datasets.load_iris`.
- Explore the features (X) and target classes (y).
- Split your data into training and testing sets (use `train_test_split` with a test size of 20% or 30%).

2. Model Training

- Initialize a `LogisticRegression` model.
- Fit the model on your training data.

3. Performance Evaluation

- Predict outcomes on your test set.
- Generate and display a Confusion Matrix to visualize misclassifications.
- Print a Classification Report (Precision, Recall, F1-Score, and Accuracy).

4. Analysis & Reflection (Required)

- *Do not just submit code.* You must write a short paragraph (5-10 sentences) analyzing your results:
- Accuracy: How well did the model perform overall?
- Errors: Look at the Confusion Matrix. Which flower species were confused with each other? Why might that be?
- Context: If this model were critical (e.g., medical diagnosis), which metric would you prioritize and why?

Deliverables

- A Jupyter Notebook (.ipynb).
- Code output showing the Confusion Matrix and Classification Report.
- Your written analysis as a comment block or markdown cell.

